

ELECENG 2EI4

Project 1

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Summary

In this project, we are tasked with designing and constructing an AC to DC power supply takes a $120V_{RMS}$, 1KHz AC input source and converts it to an expected $3V \pm 0.1V$ DC voltage that delivers 10mA. This circuit will be designed with 4 main circuit systems: Transformer, Rectification, Smoothing, and Regulation. An actual transformer will not be used, and its output will instead be simulated using an AD2.

Design

Transformer:

In this circuit the input voltage is fed through a step-down, centre tapped transformer ($n=22$) which converts the $120V_{RMS}$ source to a $5.4V_{RMS}$ output voltage. Specifically, a centre tapped transformer was used to accommodate for the specific rectifier topology used in the rectification process.

The specific output value of $5.4V_{RMS}$ was chosen to provide enough voltage for further processing steps (rectification, filtration, and regulation) to use so that the final DC output voltage is not significantly below 3V.

Rectification:

After lowering the output voltage, it must be rectified. This step will eliminate any negative voltage produced by the $5.4V_{RMS}$ AC voltage source. The specific rectification design used was the “Full Wave Centre Tapped Rectification” circuit. This design uses only 1 conducting diode at a time (rather than 2 in a full wave circuit) and hence produces a slightly higher output voltage as well as reducing experimental error. However, this design requires the use of a centre tapped transformer, as introduced in the previous step. This involves tapping the top and bottom voltages of the inductor while setting the centre of the winding to ground, which effectively makes both positive and negative halves of the 5.4V source voltage.

Assuming that each diode uses 0.7V, the output voltage of the rectifier circuit was calculated to be 3.11V.

Filter/Smoothing:

The filter circuit smooths out any variations and deviations in the output waveform. The simplest type of filter circuit is a capacitor connected parallel to the load. A series of calculations were used to determine theoretical values of both the ideal capacitance and the final DC voltage. The ideal capacitor specification must be more than $25\mu F$ and must be as close to $45.5\mu F$ as possible. Due to component restrictions, a $100\mu F$ capacitor was used. The final expected DC voltage was determined to be approximately 3.06V.

Regulator:

A regulator circuit is usually used at the end of a DC power supply circuit to eliminate and smooth out particularly noisy peak to peak voltage ripples. The simplest of these circuits is by using a Zener diode. However, the Zener diodes provided are not of sufficient specifications to be effectively used in this circuit. Hence the Zener diode regulator was not used.

Load Resistor:

The final load resistance was calculated via ohms law to be between the ranges of 290-310Ω. A resistance of 300Ω was constructed using two 150Ω resistors in series.

Calculations:

Criteria	Formulas and Calculations
AC waveforms	$V_{peak} = V_{RMS} \times \sqrt{2}$
Transformers	$\frac{V_{Primary}}{V_{Secondary}} = \frac{N_{Primary}}{N_{Secondary}} = \frac{I_{Secondary}}{I_{Primary}} = n$ $\therefore n = \frac{120V_{RMS}}{5.4V_{RMS}} \approx 22$
Rectification Diode (#1N4148)	$V_D = 0.7V$ (Assumption based on constant drop model) $V_{half,peak} = \frac{5.4V}{2} \times \sqrt{2} \approx 3.82V$ $V_{rect} = V_i - V_D = 3.82 - 0.7 = 3.11V$
Filter/Smoothing Capacitor (#CM107)	$f_r = 2f_i = 2(1KHz) = 2KHz$ $V_{ripple,max} = -0.1V \text{ to } 0.1V = 0.2V$ $V_{ripple} = \frac{I_{out}}{f_r C}$ $V_{DC} = V_{rect} - V_{ripple}$ $\therefore C_{min} \geq \frac{0.01}{2000 \times 0.2} = 25\mu F$ $\therefore V_{ripple,ideal} = 3.11 - 3.00 = 0.11$ $\therefore C_{ideal} = \frac{0.01}{2000 \times 0.11} = 45.5\mu F$ $\therefore V_{ripple} = \frac{0.01}{2000 \times 100 \times 10^{-6}} = 0.05V$ $\therefore V_{DC} = 3.11 - 0.05 = 3.06V$
Load Resistor Resistors (150Ω+150Ω)	$R = \frac{V}{I}$ $\therefore R = \frac{3 \pm 0.1V}{0.01A} = [290\Omega, 310\Omega]$

Limitations, Precautions, Design Trade-offs:

The primary limitation of the design is the absence of a regulation circuit. Since there is no Zener diode available on-hand that meet the required specifications for the design, the output DC voltage will not be tightly bounded and significant deviations/voltage spikes can occur. Another component-wise limitation is the selection of the capacitor used. A capacitor of $100\mu\text{F}$ was used as it was the only capacitor provided that best fit both capacitor specifications calculated.

Regarding safety and precautions, a 120V source was never used, and instead the lower 5.4V transformer output was simulated. This is to avoid contact with high voltages as well as the AD3 limitations. Furthermore all components chosen follow their voltage and power ratings (capacitors used can handle more than 10V, and diodes having a rating of more than 10mA)

The major design trade-off was in the transformer and rectification design used. The centre-tapped full bridge rectification design uses 1 conducting diode at a time (rather than 2 in a full wave circuit) and produces a slightly higher output voltage as well as reducing experimental error. However a centre tapped transformer is used (rather than a normal step-down transformer) which may be difficult and expensive to obtain.

Final Circuit:

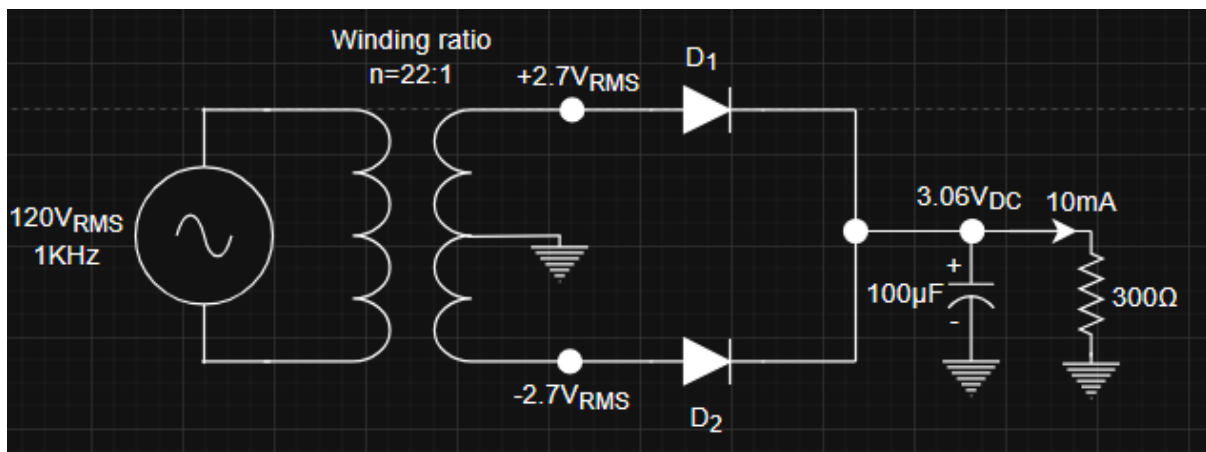


Figure 1: Schematic of Final Proposed Circuit Design

Measurement and Analysis

The final circuit was constructed using an AD2 with given components, as well as simulated on LTSpice. For the physical construction of the circuit, a transformer was not used and the outputs were simulated and fed through the rest of the circuit using the AD2 wave generator. In both the simulated and physical circuits, the voltage across the load resistor was the main measurement. The current through the resistor(s) was then derived from the measured voltage divided by the resistance.

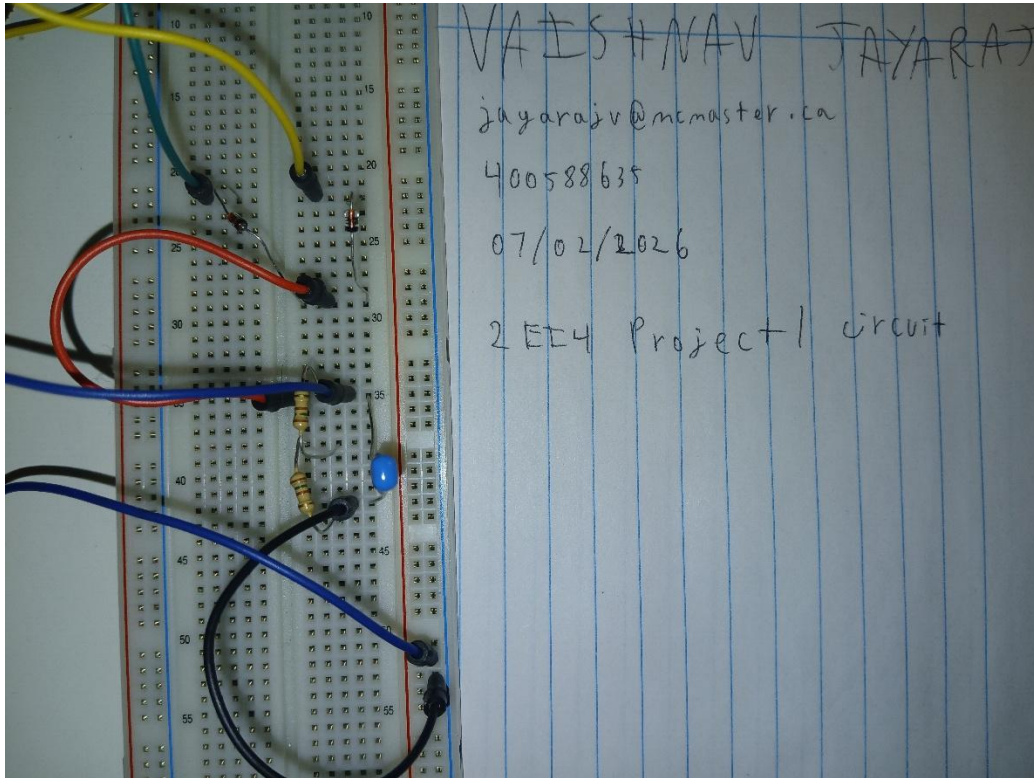


Figure 2: Image of Constructed AC to DC Power Supply Circuit

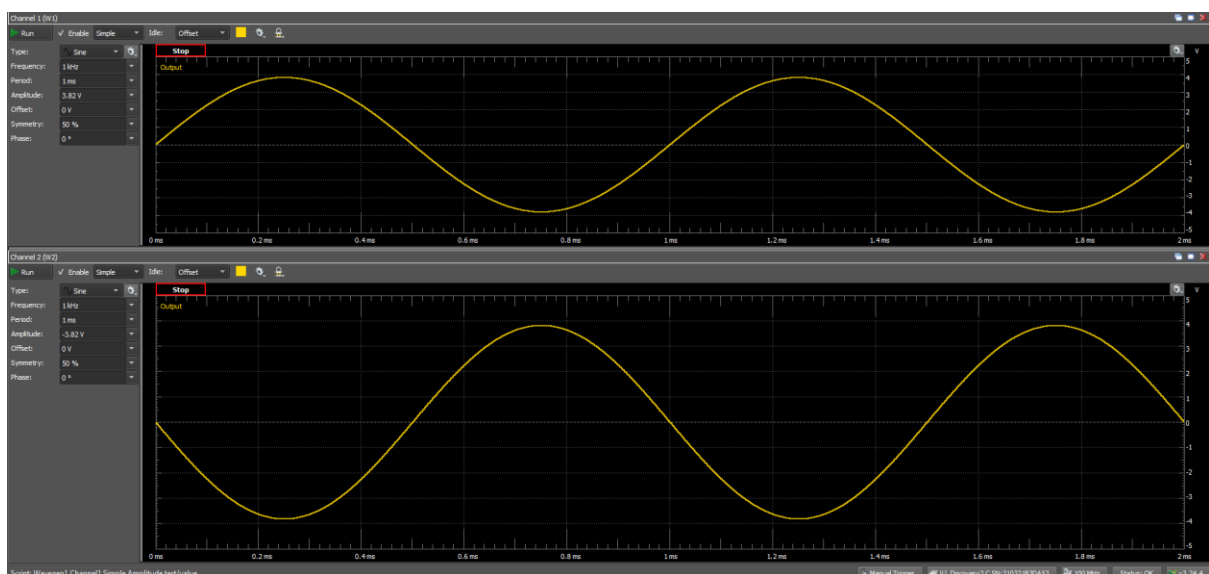


Figure 3: Voltage Inputs of Constructed AC to DC Power Supply Circuit

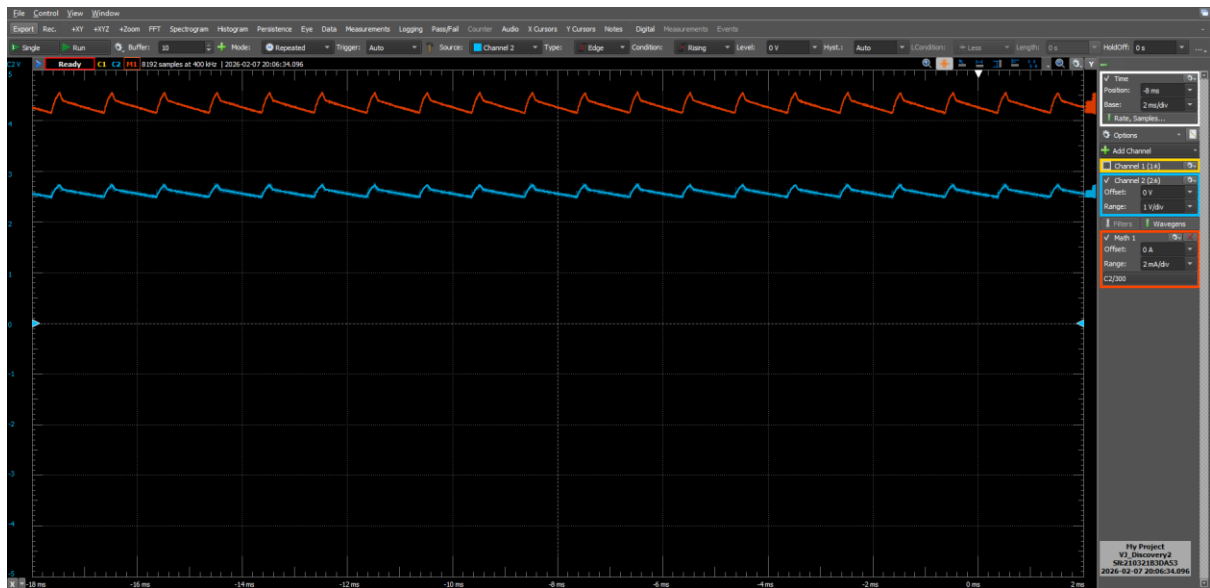


Figure 4: Voltage and Current Plots of Constructed Circuit

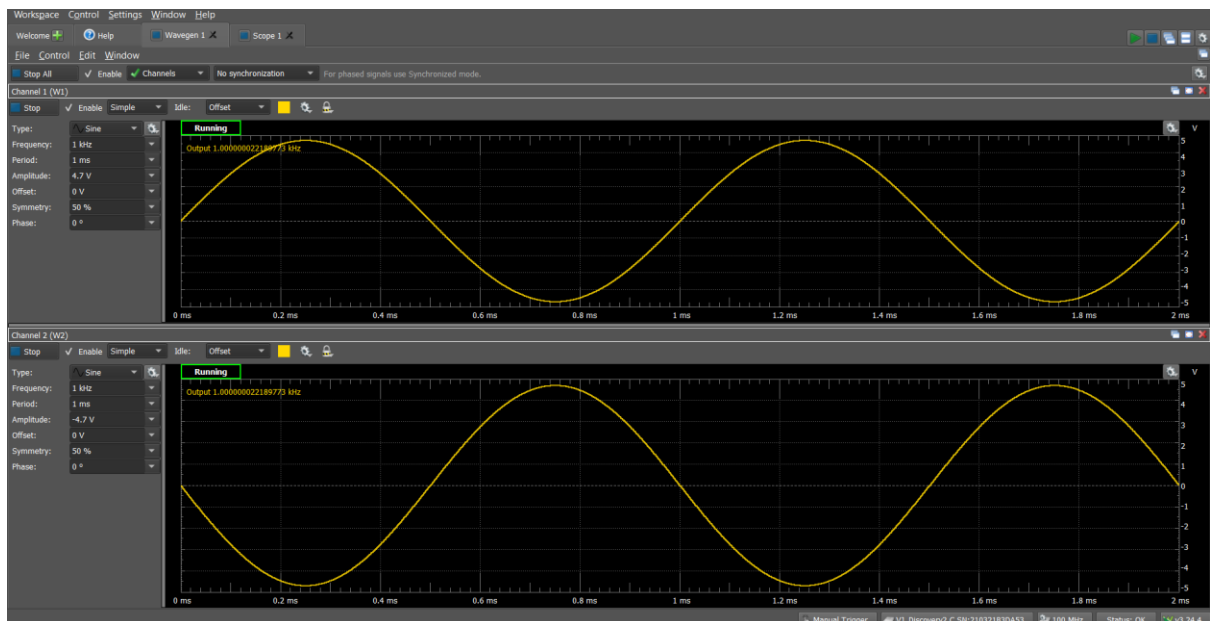


Figure 5: Corrected Voltage Inputs of Constructed AC to DC Power Supply Circuit

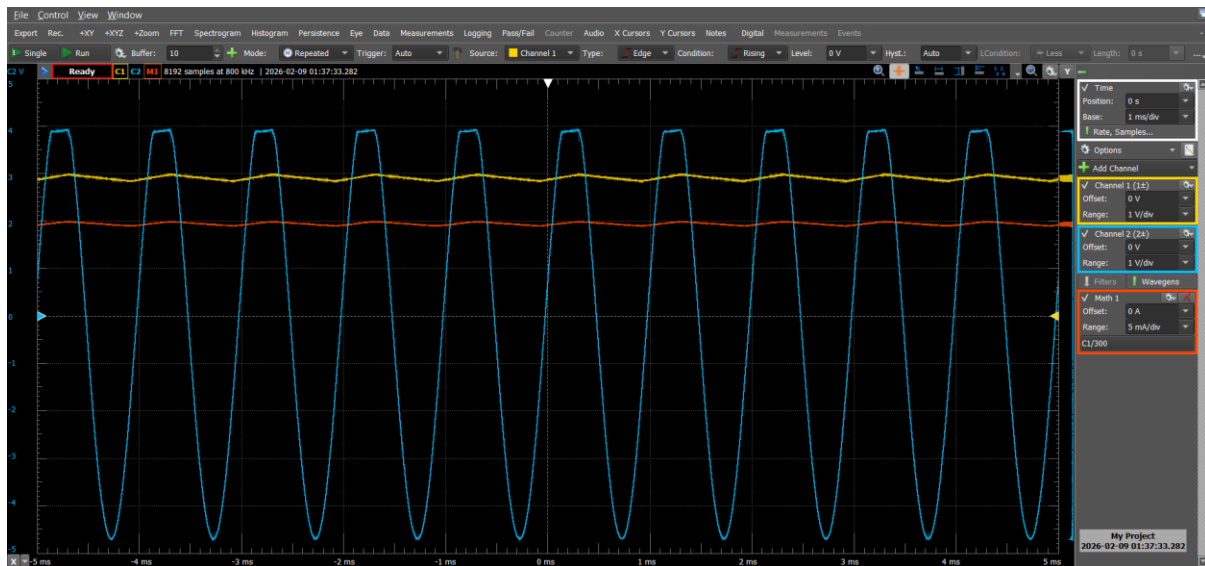


Figure 6: Corrected Voltage and Current Plots of Constructed Circuit

From the data gathered from the constructed circuit, it can be seen from the initial plots that the voltage delivered to the resistor varies from 2.5V to 2.8V. Furthermore, the current across the 300Ω resistor was shown to vary between 8.2mA to 9.2mA. The voltage produced is not within the range of $3V \pm 0.1V$ DC and has a maximum error of approximately 16% from the ideal 3V output. Similarly, the output current has a maximum error of approximately 22% to the ideal 10mA.

Due to this large error in results, a higher input voltage of approximately $\pm 4.7V$ was simulated. This led to a much better output voltage range of 3.0V to 2.9V as well as a current range of 9.0mA to 10.0mA. There was also a maximum voltage and current error of approximately 3.3% and 10% respectively.

Simulated circuit:

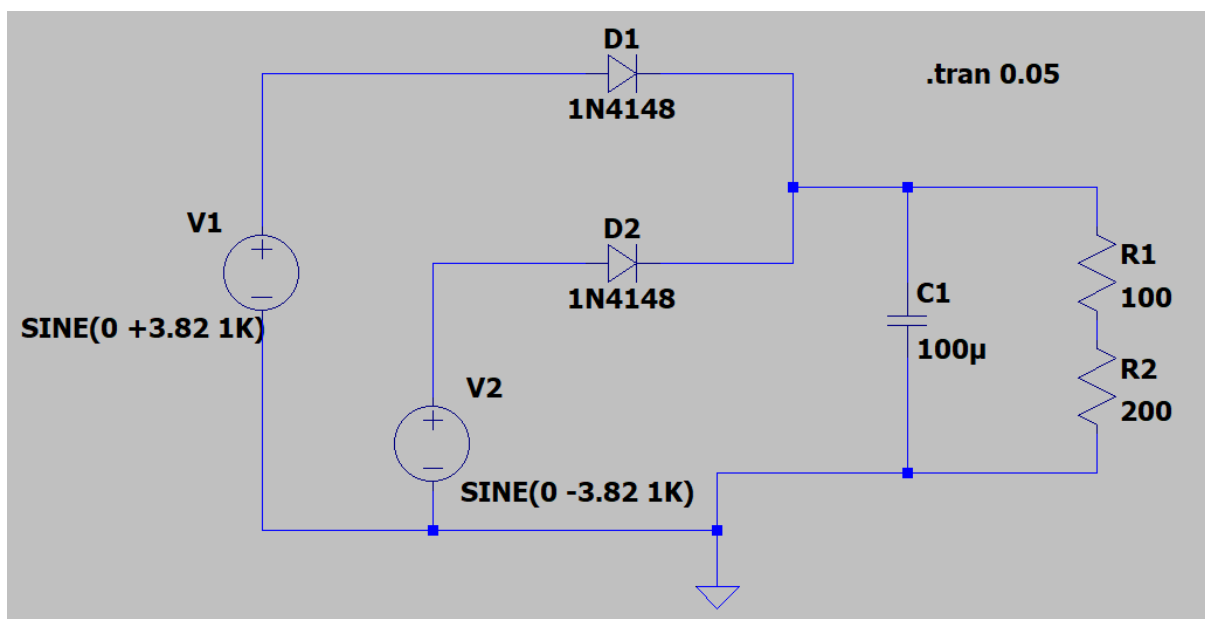
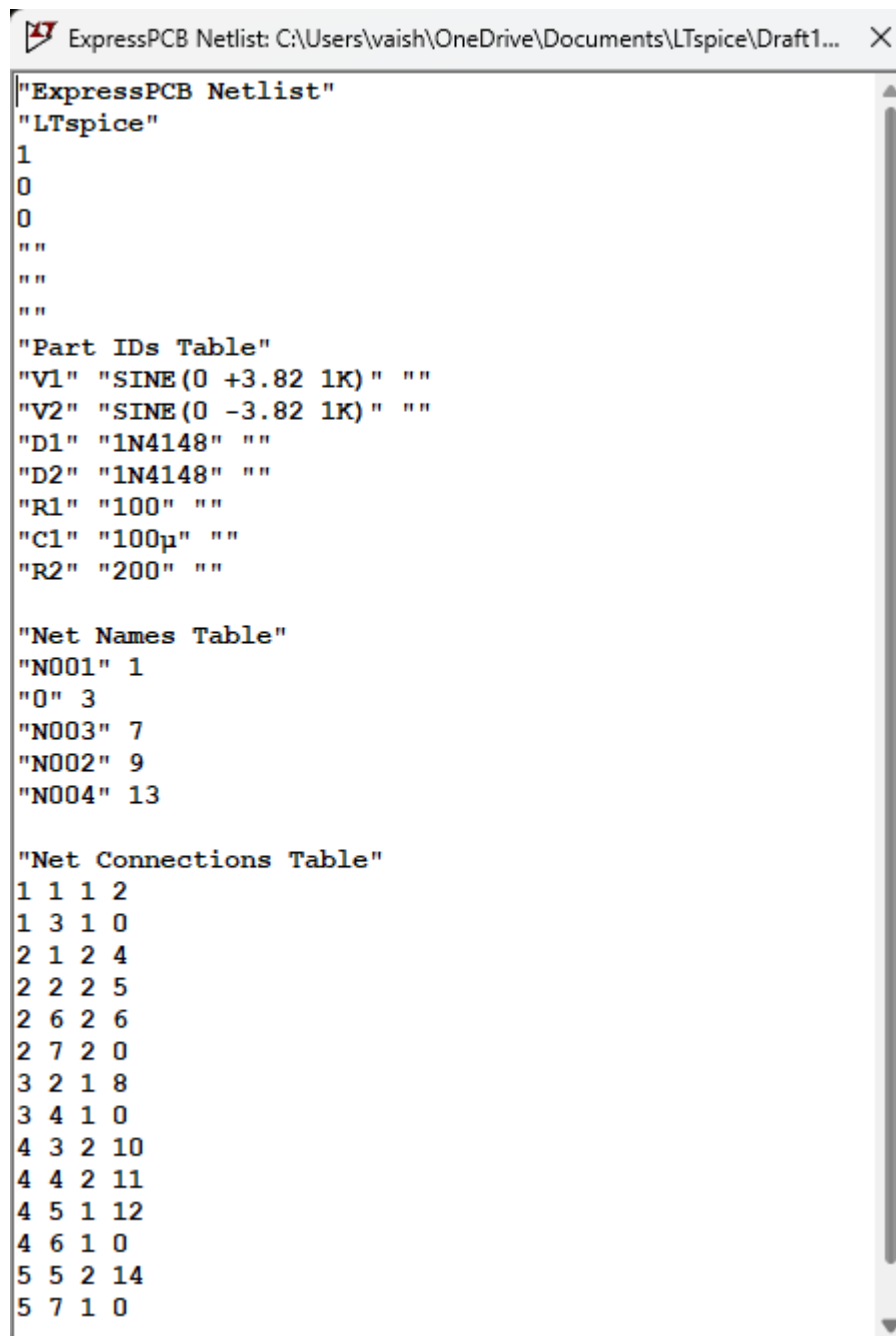


Figure 7: Schematic of Simulated Circuit

A screenshot of a software window titled "ExpressPCB Netlist: C:\Users\vaish\OneDrive\Documents\LTspice\Draft1...". The window contains a text editor with a netlist for an LTspice simulation. The netlist includes component definitions for two sine wave sources (V1, V2), two diodes (D1, D2), a resistor (R1), a capacitor (C1), and another resistor (R2). It also contains two tables: "Net Names Table" mapping net names to IDs, and "Net Connections Table" listing the connections between components and nets. The window has a standard Windows interface with a title bar, a close button (X), and a vertical scrollbar on the right.

```
"ExpressPCB Netlist"
"LTspice"
1
0
0
""
""
""
"Part IDs Table"
"V1" "SINE(0 +3.82 1K) " ""
"V2" "SINE(0 -3.82 1K) " ""
"D1" "1N4148" ""
"D2" "1N4148" ""
"R1" "100" ""
"C1" "100p" ""
"R2" "200" ""

"Net Names Table"
"N001" 1
"0" 3
"N003" 7
"N002" 9
"N004" 13

"Net Connections Table"
1 1 1 2
1 3 1 0
2 1 2 4
2 2 2 5
2 6 2 6
2 7 2 0
3 2 1 8
3 4 1 0
4 3 2 10
4 4 2 11
4 5 1 12
4 6 1 0
5 5 2 14
5 7 1 0
```

Figure 8: Netlist Data of Simulation

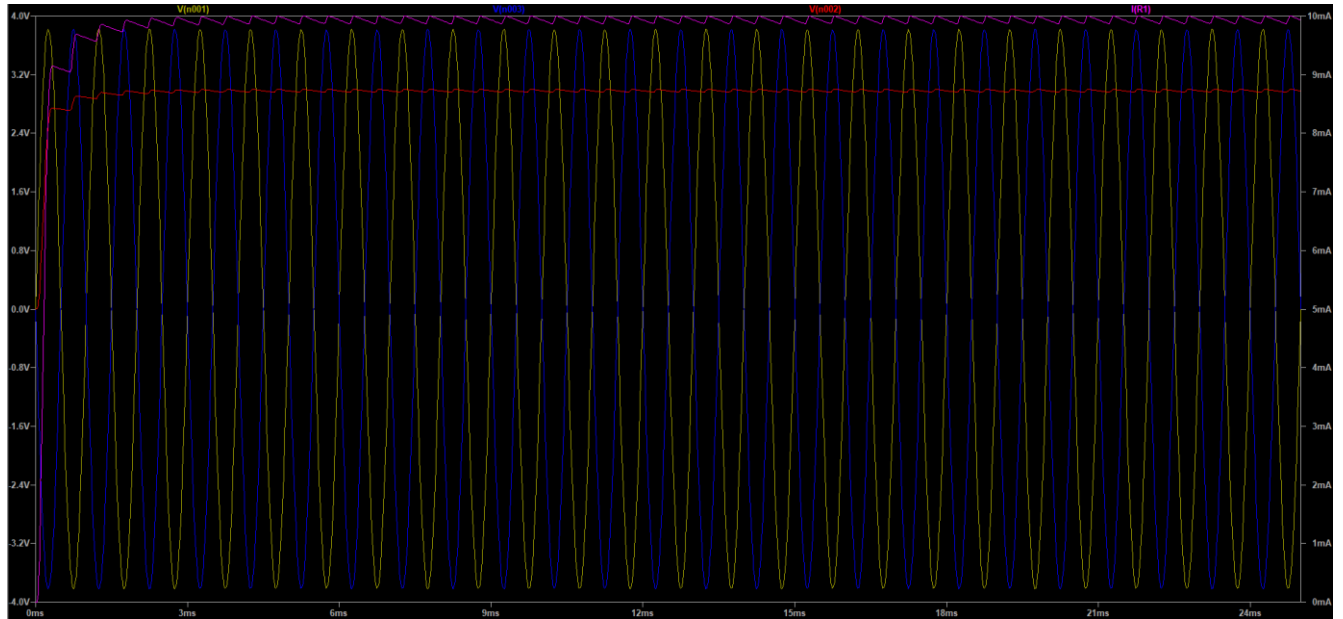


Figure 9: Final Voltage and Current Plots of Simulated Circuit

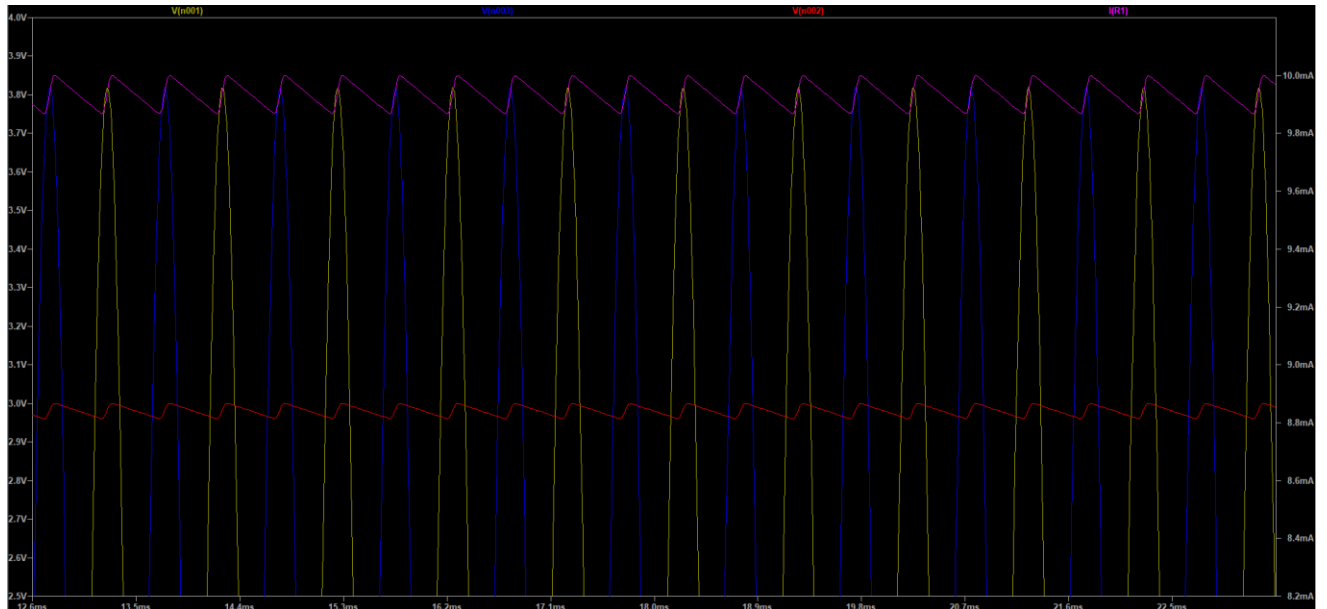


Figure 10: Final Voltage and Current Plots of Simulated Circuit (Zoomed)

From the simulations, the plots shows that that the voltage delivered to the resister varies from 2.96V to 3V. Similarly, the current across the 300 Ω resistor was shown to vary between 9.86mA to 10mA. The voltage produced by the simulation is within the range of accepted voltage ($3V \pm 0.1V$ DC) and has a maximum error of approximately 1.3% with the expected current having a max error of 1.4%.

Discussion

When comparing the output voltages produced by both the physical and simulated circuits, it can be immediately seen that they are both below the expected calculated voltage. The simulated circuit showed to be much more accurate than the physical circuit, as it has much lower margins of error. When compared to the expected and calculated final voltage and current the physical circuit had errors of both 16% and 22% respectively, whereas the simulated circuit had much better margins of error of approximately 1.3% and 1.4% respectively.

The main reason the physical circuit has a low output DC voltage and current, is due to the diodes used. From the datasheet for the diode used (1N4148), diode voltages vary from 0.7V to 1V based on the diode current. Furthermore, internal resistance of wires and environmental factors such as heat will significantly affect the value of the output voltage.

However, with these corrected considerations the transformer turns ratio can be adjusted to accommodate the increased diode voltage use. This was done experimentally by running the circuit with increasingly different input voltage levels. When the input sources on the AD2 were increased by 0.88V (+4.7V_{pp} and -4.7V_{pp}) there was a noticeable better and closer final output voltage value. The uncertainty in the voltage and current values were 3.3% and 10% respectively.

Other than the limitations of the design previously mentioned (absence of regulator, safety, and transformer accessibility) the output of the AC to DC converter circuit depends on the value of the load resistor. If there is a change in the value of the the load resistor, it will result in a different current output, which would directly affect the ripple voltage produces. This will result in higher or lower ripple spikes in the voltage which can harm the components used in the circuit as well as the load component itself. The simplest way to fix this is by implementing a regulator circuit (Zener diode or LDO regulator).

References

- [1] "23.10: Transformers," *Physics LibreTexts*, Jul. 24, 2016.
[https://phys.libretexts.org/Bookshelves/College_Physics/College_Physics_1e_\(OpenStax\)/23%3A_Electromagnetic_Induction_AC_Circuits_and_Electrical_Technologies/23.10%3A_Transformers](https://phys.libretexts.org/Bookshelves/College_Physics/College_Physics_1e_(OpenStax)/23%3A_Electromagnetic_Induction_AC_Circuits_and_Electrical_Technologies/23.10%3A_Transformers)
- [2] T. Agarwal, "Center Tapped Full Wave Rectifier : Circuit, Working & Applications," *ElProCus - Electronic Projects for Engineering Students*, Jul. 14, 2022.
<https://www.elprocus.com/center-tapped-full-wave-rectifier/>
- [3] "Small Signal Diode 1N91x, 1N4x48, FDLL914, FDLL4x48," *www.onsemi.com*.
<https://www.onsemi.com/download/data-sheet/pdf/1n914-d.pdf>
- [4] "CM107 Capacitor Ceramic 100uF 20% 6,3 volts X5R 5mm," *abra-electronics.com*.
<https://abra-electronics.com/passive-components/capacitors/ceramic-monolithic-capacitors-radial/cm107-capacitor-ceramic-100uf-20-6-3-volts-x5r-5mm.html>